

# Science

Nature Notebook  
Nature Walks & Scouting  
Biology Lessons  
Biology Labs





## About the Course

This course includes the following topic(s): Biology Lessons, Biology Labs, Nature Notebook: Grades 9-12, Nature Walks & Scouting: Grades 9-12

### About Nature Notebook: Grades 9-12

Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.

### About Nature Walks & Scouting: Grades 9-12

Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.

### About Biology Lessons

Alveary High School Biology guides students through all major topics in the discipline. Based on the second edition of Novare General Biology, the Alveary course incorporates living engagement and special attention to citizenship, problem-solving, and communication skills for a complete Charlotte Mason science course. Members who already have the first edition of the text should be sure to keep their lesson plans for future use!

### About Biology Labs

Labs are an essential part of science in which students engage with the Things they are reading about and practice the scientific method. Labs for this course are integrated into the lessons to facilitate adequate time for more involved activities and to better coordinate with the lessons.



## Placement & Combining Tips

### Nature Notebook: Grades 9-12

Learners may be combined and follow their own interests.

### Nature Walks & Scouting: Grades 9-12

Learners may follow their own interests or follow the plan of their local scouting troop or natural history club.

### Biology Lessons

There are no specific prerequisites or concurrents for this course, but there are a number of complex ideas that are difficult without some prior experience. A course with some exposure to basic chemistry is recommended before enrollment in Biology.



## Scheduling

| GRADE  | SCHEDULE INFO.                                              | BOOKS                                                                                                                 |
|--------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| 9-12   | Nature Notebook: Grades 9-12<br>1+ time/week 20 min+        |                                                                                                                       |
| 9-12   | Nature Walks & Scouting: Grades 9-12<br>1 time/week 30 min+ |                                                                                                                       |
| 10-12+ | Biology Lessons<br>5 times/week 45 min                      | The Biology Coloring Book<br>The Spirit of the Rainforest<br>General Biology, 2nd Edition<br>The Fool and the Heretic |
| 10-12+ | Biology Labs<br>1 time/week 60 min                          | Alveary Biology Lab Book                                                                                              |

## Sample Weekly View

| Day 1                   | Day 2                                                      | Day 3           | Day 4                                              | Day 5                           |
|-------------------------|------------------------------------------------------------|-----------------|----------------------------------------------------|---------------------------------|
| <b>Science: Biology</b> |                                                            |                 |                                                    |                                 |
| Biology Lessons         | Biology Lessons<br>Nature Walks &<br>Scouting: Grades 9-12 | Biology Lessons | Biology Lessons<br>Nature Notebook:<br>Grades 9-12 | Biology Lessons<br>Biology Labs |



## Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

**LINKS:** Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

### Biology Lessons

☐ Carve out time to continue or establish the regular habit of spending time in nature, including the use of a nature journal, as appropriate.

☐ Obtain materials from the supply lists.

☐ Plan for discussion and engagement. Teachers who do not feel that they can discuss the subject of this course with students should plan to either pre-read the material or complete the course alongside their students. Expertise is not necessary, but discussion is as important in math and science as in history and literature.

☐ Labs/activities are presented in a fully integrated state in which they are connected to the lessons. To use them in this way, it is helpful for the lab to be scheduled after the 5th lesson. If your lab time is earlier in the week and you will find it difficult to be flexible, plan to delay the start of the lesson schedule so that lab day occurs after the 5th lesson. For example, if your lab time is on Tuesday, you might begin lesson 1 on Wednesday of the first week of school rather than on Monday. Those wanting to use the labs/activities on a more flexible schedule can do so, but should be ready and able to adjust lesson connections.

☐ Select a science book from the Alveary bookshelf for personal reading time, as appropriate.

## Term Prep & Teacher Tips

### Biology Lessons

☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:

- any living plant specimen found nearby
- any non-living specimen
- red onion (Term 1)
- kitchen knife
- distilled or deionized water
- paper towels
- lamp or bright light
- strawberries (Term 1)
- salt
- a variety of colors of small Lego bricks
- computer access
- stopwatch, such as those found on most electronic devices
- cereal O's
- small bread cubes or mini marshmallows
- chopsticks
- fork

## Reminders

### Biology Lessons

☐ Note that Lab in week 5 calls for red onions and Lab in week 8 for strawberries. Make a note in your calendar if you will need to purchase these closer to the appropriate time. Refer to the

lab book for more details.

□ Note that the Keys (provided in Quick Links) are for the 1st edition. The overall content in the two editions is very similar, but the numbering of chapters is slightly different, especially as learners reach Term 3. When using the Keys, teachers should attend to the topics rather than relying on the numbering.



## Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

### Science: Biology

#### Biology Lessons



The Biology Coloring Book



The Spirit of the Rainforest



General Biology, 2nd Edition



The Fool and the Heretic

#### Biology Labs



Alveary Biology Lab Book



## Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

∞ [View Supply List Details](#)

### Science: Biology



Alveary High School Biology Kit



Elodea



Household Items - Science: Biology



Prepared Microscope Slides



Frog Dissection Specimen



Student Microscope

#### Biology Labs



Slides and Coverslips



Safety Goggles



## Quick Links

### Science: Biology

- ∞ [Extra Helpings](#)
- ∞ [Biology Lab Book](#)
- ∞ [AmScope Original Student Slide Key](#)

### Biology Lessons

- ∞ [All Keys and Sample Answers \(1st edition, see reminder in header\)](#)
- ∞ [Homework Help for HS Science Students](#)
- ∞ [Contact Us for Teachers](#)
- ∞ [Alveary Bookshelf](#)
- ∞ [John Muir Laws • Nature Journaling Resources](#)
- ∞ [Seek app from iNaturalist](#)
- ∞ [SkyView Lite for iOS](#)
- ∞ [SkyView Lite for Android](#)
- ∞ [Lab Notebook Examples](#)

Click THIS text or scan the QR code for links.



# Science: Biology

## How To Approach



### Introduce

- Learners begin each lesson with their existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to consider what learners think about it, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.
- Take note of the chapter or section heading, and consider how the day's topic connects back to prior topic(s).
- Look back as needed. The outline of the book and even of the chapter is helpful in drawing out and connecting ideas. Learners practice making these connections as they proceed through the book.



### Read

- Read or do as instructed in the lessons. Teachers should note any Teacher Tips provided.
- Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.
- As they read, learners record ideas in a notebook or binder using outlines, diagrams, graphic organizers, or other methods (or a combination of methods) that suit them. These recordings can be a helpful mechanism for remembering or a mini-narration to support understanding.
- If learners do not understand a word or concept, do not worry. Try reading over the passage again, studying a picture or diagram, connecting the idea to something from real life, or practicing chapter exercises. The lab book further supports major concepts from the text.



### Narrate

- Process the ideas of the lesson by explaining a concept, describing an object, retelling events, etc. Consider what the point of the lesson is before deciding what to do with it.
- Learners may use words, pictures, models, graphics, etc., to process and convey ideas in their own way.



### Discuss

- Consider together any thoughts, confusion, or concerns about the passage.
- Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher.
- Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.
- Notice if there are any dates to keep for the Book of Centuries or quotes for the Commonplace Book.



## Connect

- Follow any extra links, examine any sidebars in the text, look at pictures, etc, depending on learner needs and interest.
-

# Science: Biology

## How To Approach Labs



### Introduce

- Before beginning, be sure that the basic rules of lab safety (as listed in the lab manual) are understood and obeyed.
- Every science lab has the same flow, which follows the scientific method and is guided by the lab book.
- Learners begin with an introduction in their lab book. How does this lab activity relate to what they have learned so far, as well as any previous experience? What will they learn from the lab? This is how hypotheses are formed.
- If learners are unsure what to do with the Introduction, it can help to dialogue with someone as they work through the ideas. Check the Chemistry Companion in the Quick Links if a video tutorial might be helpful. This year is a pilot for this video series, so videos will be added to the Quick Link throughout the year.
- Once they have had a chance to think, learners compose a prelab narration to put these introductory ideas and hypotheses into their lab notebooks. These prelab narrations may seem short and even incomplete if learners are new to keeping a lab notebook, but that is okay.
- If learners have difficulty or are easily frustrated, then consider allowing a digital notebook for typing, a scribe, or use of assistive technology, as appropriate.
- After they complete their prelab narration, learners should collect the listed materials, being sure to let teachers know if something is missing or to remind the teacher if something needs to be purchased at the store.



### Lab Procedure

- Learners complete the procedure as instructed in the lab book.
- The lab gives instructions for using the lab notebook to create tables and diagrams.



### Analysis & Conclusions

- The last step in the lab is to analyze the data and observations and draw conclusions from them. This postlab narration is prompted in the lab manual. Similar to the prelab narration, the concluding or postlab narration is a chance to think about and put into words. This time, learners are considering what they learned from the lab and what they could learn more about, if they were to continue.
- Teachers should engage learners with this reflection by reviewing their lab notebooks with them, discussing the science used in the lab, and demonstrating curiosity about the lab themselves.
- Depending on the interest of the learner and the priorities of the teacher, the learner might be encouraged to spend more time on those ideas of what more they could learn or it might be time to move on. Either way, it is an important part of the scientific method to reflect on what we could or would do next - our practice should help to clarify our thinking and teach that there is always more to be learned.



## Term 1

### WEEK 1 45m Biology Lessons - Lesson 1

Materials: General Biology

PREP: Be sure you have read the Course Notes and Planning and Prep.

#### → INTRODUCTION

In General Biology, much of your previous experience in science comes together. You will notice some familiar ideas and will explore many of them with greater depth than ever before. As you work through the course, note the outline of ideas in the text and the main ideas in each section, and record any definitions and important points. You should take a few minutes to review your notes at least weekly throughout the year.

#### → READ, NARRATE, & DISCUSS

Read the introduction to Ch.1 and the Objectives for Section 1.1. Then read to the end of Section 1.1, noting important terms and their definitions, copying any helpful diagrams, and recording interesting or important points. You will do this as you continue throughout the course. Note that the author of this series uses the term "Cycle of Scientific Enterprise" to refer to the process that is the Scientific Method (1.1.3). If you have completed previous levels of Alveary Science and have a working understanding of the Scientific Method from lab, you may quickly skim section 1.1.3 or simply review the process in Figure 1.6.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 1 45m Biology Lessons - Lesson 2

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 1.2 at the beginning of the chapter. Then read Section 1.2.1 - 1.2.2.

### WEEK 1 45m Biology Lessons - Lesson 3

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Section 1.2.3 - 1.2.7

### WEEK 1 45m Biology Lessons - Lesson 4

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 1.3 at the beginning of the chapter. Then, finish Ch.1 by reading section 1.3. Use any time remaining to begin the review exercises at the end of the chapter. You may answer these orally, in writing, or with pictures. If unsure about any of them, refer back to the appropriate section to reread or find the answer.

### WEEK 1 45m Biology Lessons - Lesson 5

Materials: General Biology

#### → PRACTICE

Continue working through the review exercises at the end of Ch.1.



## Term 1

### WEEK 1 60m Biology Lessons - Lesson 6

*Observing Living and Nonliving Things*

Materials: High School Biology Lab Book and materials listed within, and The Biology Coloring Book

PREP: Read Student/Teacher Tip

#### → LAB DAY

Complete Lab 1. Color plate 1 in your coloring book, as time permits.

#### ★ TEACHER/STUDENT TIP

**Note that there are 3 activity days next week: Lessons 7-9.**

Two of these require outdoor work. You may move Week 2 to accommodate the weather, as needed. If you need to adjust for a rainy day in your forecast for the week, then Lesson 7 can be moved to another day of the week.

### WEEK 2 45m Biology Lessons - Lesson 7

Materials: Article link

#### → READ, NARRATE, & DISCUSS

∞ Article Link: In the Laboratory

Study any natural object, recording what you see. When you think you have noticed all that you can, try to observe more. What else can you see? You may use field guides, handbooks, etc., to aid your study today, if desired.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 2 45m Biology Lessons - Lesson 8

Materials: Article links

#### → INTRO

A BioBlitz is a biological inventory where scientists find and identify as many species as possible in a specific area during a specific time. If you'd like to learn more about it, you can read the National Geographic article linked. A bigger question is why scientists would take such an inventory, however. To understand that, read the article on current events from Science News Explores.

∞ Article Link: BioBlitz

#### → READ, NARRATE, & DISCUSS

Read the article entitled, "Some Spikes in Malaria Cases May Be Due to Amphibian Die-off" from Science News Explores

∞ Article Link: Some spikes in malaria

#### → FIELDWORK

Your BioBlitz will last for 1 day, so choose your area carefully based on how many scientists will be keeping inventory and what you plan to record. You may inventory all species or focus on only one type of object (e.g., plants, insects, birds, etc.). You will only briefly observe, identify, and document each object in your lab notebook - you will not journal on individuals as in Lab 1. Today, decide what you will inventory and plan how you will collect your BioBlitz data. Collect any special materials or equipment that you might need, such as a magnifier or field guide. Then, begin by writing an introductory narration in your notebook and getting an overview of the area. You might want to sketch a landscape or make a map of the area, noting any landmarks.

### WEEK 2 45m Biology Lessons - Lesson 9

Materials: anything needed for data collection

#### → FIELDWORK

Collect your BioBlitz data.



## Term 1

### WEEK 2 45m Biology Lessons - Lesson 10

Materials: graph paper or spreadsheet

#### → DATA ANALYSIS

Analyze your BioBlitz data. Do you want to make a table or chart of the data? How can you best represent what you noticed? Do you think your data would be different in a different location or time of year? How could you find out?

### WEEK 2 45m Biology Lessons - Lesson 11

Materials: Links

#### → REPORT WRITING

Write a report indicating the results of your BioBlitz. Your report should have an Introduction explaining the purpose of the field study, a Procedure explaining your location and time, any methods (e.g., sample collection, identification, etc.), and any special equipment used in your study, Results indicating your findings, and Analysis and Conclusions explaining interesting observations, introducing any questions, and sharing whatever conclusions you came to. There is an example in the link. The article links we started with are also provided if you want to refer back to them. You may work on this report during Lesson 12, as well.

∞ Link: Sample Lab Report

∞ Article Link: BioBlitz

∞ Article Link: Some spikes in malaria

### WEEK 2 60m Biology Lessons - Lesson 12

Materials: Report from previous lesson

#### → REPORT WRITING

Complete your report. Review and edit it to make improvements before turning it in. While editing, be sure you have used complete sentences and avoid using the first person. In technical writing, such as this, the sentence "I tied the string to the end of the meter stick" should be rephrased, "The string was tied to the end of the meter stick."

### WEEK 3 45m Biology Lessons - Lesson 13

Materials: General Biology

#### → NOTE

If you completed Introductory Chemistry, you may find that much of Ch.2 is a review with some application to biological systems. If you have not yet completed Introductory Chemistry, you may need to take this chapter more slowly.

#### → READ, NARRATE, & DISCUSS

Read the introduction to Ch.2 and the Objectives for Section 2.1. Note important terms and their definitions, copy any helpful diagrams, and record any interesting or important points.

Then read sections 2.1.1 - 2.1.2, working the examples in the text. Always work the examples in the text as you read because these help to ensure understanding.

If you have time remaining, begin to work through the Exercises at the end of the chapter for 2.1. For general practice and review, note that

#### • PLAN WEEKLY

nature walk - record observations

science free read



## Term 1

coloring plates 4-21 cover the material in Ch.2. You can use any plates like these that are not scheduled when you need some extra practice or review.

### WEEK 3 45m Biology Lessons - Lesson 14

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Section 2.1.3. Read and copy any diagrams that you think are helpful to your understanding.

Then, continue work on the Exercises for Section 2.1, as time permits.

#### ★ STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

### WEEK 3 45m Biology Lessons - Lesson 15

Materials: General Biology

PREP: Read Teacher Tip

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 2.2 at the beginning of the chapter. Read 2.2 and copy any diagrams that you think are helpful to your understanding.

Then work on the Exercises for Section 2.2, as time permits.

### WEEK 3 45m Biology Lessons - Lesson 16

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 2.3 at the beginning of the chapter. Read sections 2.3.1 - 2.3.2

Then work on the Exercises for Section 2.3, as time permits.

#### ★ STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

### WEEK 3 45m Biology Lessons - Lesson 17

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Finish Section 2.3

Then work on the Exercises for Section 2.3, as time permits.

### WEEK 3 60m Biology Lessons - Lesson 18

*Collecting Properties of Water*

Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Begin Lab 2.

#### ★ STUDENT/TEACHER NOTE

Lab 2 will be completed over two lessons. (Lessons 18 and 24)

### WEEK 4 45m Biology Lessons - Lesson 19

Materials: General Biology

#### → PRACTICE

Continue working through the review exercises at the end of Ch.2.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read



## Term 1

### WEEK 4 45m Biology Lessons - Lesson 20

Materials: General Biology

#### → REVIEW

Remember to briefly look back over your notes to help recall what you have learned so far.

#### → READ

Read the introduction to Ch.3 and the Objectives for Section 3.1 at the beginning of the chapter. Note important terms and their definitions, copy any helpful diagrams, and record any interesting or important points.

Then, read Section 3.1.1.

If you have time remaining, you may begin to work through the Exercises for Ch.3

#### • STUDENT TIP

There are a lot of pieces and parts with a lot of new names in this chapter. Focus on understanding how they work and how they fit together rather than remembering the names. The names will come in time.

### WEEK 4 45m Biology Lessons - Lesson 21

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Sections 3.1.2 - 3.2.1, referring to the Objectives for Section 3.2 before beginning that section.

### WEEK 4 45m Biology Lessons - Lesson 22

Materials: General Biology and Coloring Book

#### → READ, NARRATE, & DISCUSS

Read to the end of the section on the Endomembrane System.

Begin plates 30-32 as time permits. You will have more time for these later.

#### ★ STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

### WEEK 4 45m Biology Lessons - Lesson 23

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Read to the end of section 3.2

#### ★ STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

### WEEK 4 60m Biology Lessons - Lesson 24

*Collecting Properties of Water*

Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Complete Lab 2.

### WEEK 5 45m Biology Lessons - Lesson 25

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 3.3 at the beginning of the chapter. Then read Section 3.3. Copy any diagrams that are helpful to you.

Then continue to work on Ch.3 Exercises, as time permits.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read



## Term 1

### WEEK 5 45m Biology Lessons - Lesson 26

Materials: Coloring Book

#### → COLOR

Plates 30-32 and 36

### WEEK 5 45m Biology Lessons - Lesson 27

Materials: General Biology

#### → PRACTICE & REVIEW

Continue working through the review exercises at the end of Ch.3, reviewing as needed.

### WEEK 5 45m Biology Lessons - Lesson 28

Materials: General Biology

#### → PRACTICE & REVIEW

Continue working through the review exercises at the end of Ch.3, reviewing as needed.

### WEEK 5 45m Biology Lessons - Lesson 29

Materials: General Biology and Coloring Book

#### → REVIEW

This is a good point in the book to do a cumulative review of what you have learned so far. Look back over your notes/narrations and the chapter objectives. Review or use unassigned coloring plates for anything difficult to recall.

### WEEK 5 60m Biology Lessons - Lesson 30

*Observing Cells and Passive Transport*

Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Complete Lab 3.

### WEEK 6 45m Biology Lessons - Lesson 31

Materials: General Biology

#### → READ

Read the introduction to Ch.4 and the Objectives for Section 4.1 at the beginning of the chapter. Note important terms and their definitions, copy any helpful diagrams, and record any interesting or important points.

Then read Section 4.1

If you have time remaining, you may begin to work through the Exercises for Ch.4.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 6 45m Biology Lessons - Lesson 32

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 4.2 at the beginning of the chapter. Then read Sections 4.2.1-4.2.2 through Oxidation of Pyruvate.

#### • STUDENT TIP

If the processes and figures in this section (and the rest of Ch.4) feel overwhelming, break them into smaller parts and remember that we are trying to understand the



## Term 1

|               |                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                             |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |                                                                                                                                                                                                                                                                                                                            | big ideas rather than memorize all the details.                                                                                                                                                                                                                                                                                                             |
| <b>WEEK 6</b> | <b>45m Biology Lessons - Lesson 33</b><br><div>Materials: General Biology</div><br><b>→ READ, NARRATE, &amp; DISCUSS</b><br>Finish section 4.2.2                                                                                                                                                                           | <b>★ STUDENT REMINDER</b><br>Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.                                                                                                                                                                                                               |
| <b>WEEK 6</b> | <b>45m Biology Lessons - Lesson 34</b><br><div>Materials: General Biology, High School Biology Lab Book, and Coloring Book</div><br><b>→ LAB PREP</b><br>Complete step 1 of Lab 4 Procedure.<br><br><b>→ READ, NARRATE, &amp; DISCUSS</b><br>Section 4.2.3<br><br>Then color plates 53-55 as time permits.                 | <b>★ STUDENT/TEACHER NOTE</b><br>Lessons 34-36<br>In Lab 4, students will complete a step of the lab outside of their normal lab day during Lessons 34 and 35. In addition, they will also complete coursework. Be sure to remind your student to manage their time in those lessons. Lessons 34 and 35 are 45 minutes each, while lesson 36 is 60 minutes. |
| <b>WEEK 6</b> | <b>45m Biology Lessons - Lesson 35</b><br><div>Materials: General Biology and High School Biology Lab Book</div><br><b>→ LAB PREP</b><br>Complete step 2 of Lab 4 Procedure.<br><br><b>→ PRACTICE &amp; REVIEW</b><br>Work through the review exercises at the end of Ch.4, reviewing as needed.                           | <b>★ STUDENT REMINDER</b><br>Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!                                                                                                                                                                                                                  |
| <b>WEEK 6</b> | <b>60m Biology Lessons - Lesson 36</b><br><i>Measuring Respiration in Plants</i><br><div>Materials: High School Biology Lab Book and materials listed within</div><br><b>→ LAB DAY</b><br>Continue Lab 4 by reading the Introduction. Note that you will need a few minutes to complete the final observation in 24 hours. |                                                                                                                                                                                                                                                                                                                                                             |
| <b>WEEK 7</b> | <b>45m Biology Lessons - Lesson 37</b><br><div>Materials: General Biology</div><br><b>→ READ, NARRATE, &amp; DISCUSS</b><br>Refer to the Objectives for Section 4.3 at the beginning of the chapter. Read to the end of section 4.3.2.                                                                                     | <b>• PLAN WEEKLY</b><br><div>nature walk - record observations</div><br><div>science free read</div>                                                                                                                                                                                                                                                        |
| <b>WEEK 7</b> | <b>45m Biology Lessons - Lesson 38</b><br><div>Materials: General Biology</div><br><b>→ READ, NARRATE, &amp; DISCUSS</b><br>Sections 4.3.3-4.3.5                                                                                                                                                                           | <b>★ STUDENT REMINDER</b><br>Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.                                                                                                                                                                                                               |



## Term 1

### WEEK 7 45m Biology Lessons - Lesson 39

Materials: Coloring Book

#### → COLOR

Continue coloring any unfinished pages. Color plates 56-57.

### WEEK 7 45m Biology Lessons - Lesson 40

Materials: General Biology

#### → PRACTICE & REVIEW

Work through the review exercises at the end of Ch.4.

### WEEK 7 45m Biology Lessons - Lesson 41

Materials: General Biology

#### → PRACTICE & REVIEW

Work through the review exercises at the end of Ch.4.

### WEEK 7 60m Biology Lessons - Lesson 42

*Measuring Photosynthesis in Plants*

Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Complete Lab 5.

### WEEK 8 45m Biology Lessons - Lesson 43

Materials: High School Biology Lab Book

#### → REPORT WRITING

Write a lab report for the photosynthesis lab. This is simply a formal report based on what is already in your lab notebook. Just like your last report, you should include an Introduction (a more polished version of your introductory narration), Methods (this is your procedure or any important information that someone repeating your experiment would need to know); Results (this is your data, including any tables, graphs, or photographs that you want to provide); a Discussion or Data Analysis; and a Conclusion. The last two sections are a more polished version of your analysis and concluding narration.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 8 45m Biology Lessons - Lesson 44

Materials: High School Biology Lab Book

#### → REPORT WRITING

Complete your report. Review and edit the lab report before turning it in. Use complete sentences, good grammar, 3rd person voice, correct tense, significant figures, and scientific notation, as appropriate. Read the report as if you were another scientist who is unfamiliar with the project. Does the report make sense? Can you repeat the project using the lab report? If not, edit the report to make it so.

### WEEK 8 45m Biology Lessons - Lesson 45

Materials: General Biology

#### → REVIEW

Complete any unfinished work. Then take some time to do a cumulative



## Term 1

review of what you have learned so far. Look back over your notes and the chapter objectives. Review anything difficult to recall.

### WEEK 8 45m Biology Lessons - Lesson 46

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Read the Introduction to Ch.5 and the Objectives for Section 5.1 at the beginning of the chapter. Note important terms and their definitions, copy any helpful diagrams, and record any interesting or important points.

Then read Section 5.1.1

If you have time remaining, you may begin to work through the Exercises for Ch.5 and/or color plate 59.

#### ★ STUDENT REMINDER

Remember to break any overwhelming processes or figures into smaller parts or steps and focus on getting at the ideas rather than memorizing.

### WEEK 8 45m Biology Lessons - Lesson 47

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Sections 5.1.2-5.2.1, referring to the Objectives for Section 5.2 before beginning that section.

#### ★ STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

### WEEK 8 60m Biology Lessons - Lesson 48

*Extracting DNA*

Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Complete Lab 6.

### WEEK 9 45m Biology Lessons - Lesson 49

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Read to the end of the section on Transcription, referring to the Objectives at the start of the chapter if you begin a new section in your edition.

#### ● PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 9 45m Biology Lessons - Lesson 50

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Read the section on Translation.

### WEEK 9 45m Biology Lessons - Lesson 51

Materials: General Biology and Coloring Book

#### → READ, NARRATE, & DISCUSS

Read the section on Post-translational Modification and Regulation.

Then color plates 82, 84, and 85 as time permits. You will have more time for these next week.

#### ★ STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!



## Term 1

### WEEK 9 45m Biology Lessons - Lesson 52

Materials: General Biology

#### → PRACTICE & REVIEW

Work through the review exercises at the end of Ch.5.

#### ★ STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

### WEEK 9 45m Biology Lessons - Lesson 53

*Decoding DNA*

Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Begin Lab 7.

### WEEK 9 60m Biology Lessons - Lesson 54

*Decoding DNA*

Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Complete Lab 7.

### WEEK 10 45m Biology Lessons - Lesson 55

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for the Section at the beginning of the chapter.  
Read to the end of the section on Mitosis and Cytokinesis.

#### ● PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 10 45m Biology Lessons - Lesson 56

Materials: General Biology and Coloring Book

#### → READ, NARRATE, & DISCUSS

Read the section on Regulation of the Cell Cycle.

Then color plates 65 and 83 as time permits. You will have more time for these tomorrow.

#### ★ STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

### WEEK 10 45m Biology Lessons - Lesson 57

Materials: Coloring Book

#### → COLOR

Continue coloring plates.

### WEEK 10 45m Biology Lessons - Lesson 58

Materials: General Biology

#### → PRACTICE & REVIEW

Work through the review exercises at the end of Ch.5.



## Term 1

### WEEK 10 ☐ 45m Biology Lessons - Lesson 59

☐ Materials: General Biology

#### → PRACTICE & REVIEW

Work through the review exercises at the end of Ch.5.

### WEEK 10 ☐ 60m Biology Lessons - Lesson 60

*Observing Mitosis*

☐ Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Complete Lab 8. Then, complete any unfinished lab activities and notebook work, or read over the Citizen Science project for next term and discuss with your teacher whether you need to procure (gather) materials.

### WEEK 11 ☐ 45m Biology Lessons - Lesson 61

☐ Materials: General Biology

#### → REVIEW & CATCH-UP

Work through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 11 ☐ 45m Biology Lessons - Lesson 62

☐ Materials: General Biology

PREP: Read Teacher Tip

#### → REVIEW & CATCH-UP

Work through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments.

### WEEK 11 ☐ 45m Biology Lessons - Lesson 63

☐ Materials: General Biology

#### → REVIEW & CATCH-UP

Work through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments.

### WEEK 11 ☐ 45m Biology Lessons - Lesson 64

☐ Materials: General Biology

#### → REVIEW & CATCH-UP

Work through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments.

### WEEK 11 ☐ 45m Biology Lessons - Lesson 65

☐ Materials: General Biology

#### → REVIEW & CATCH-UP

Work through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments.



## Term 1

|                                                                                                                                                                                    |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div><div>WEEK 11</div><div><div><div></div></div> 60m Biology Lessons - Lesson 66</div></div>                                                                                     |  |
| <div><div><div></div> Materials: General Biology</div></div>                                                                                                                       |  |
| <div><div>→ REVIEW &amp; CATCH-UP</div><div>Work through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments.</div></div> |  |
| <div><div>WEEK 12</div><div><div><div></div></div> 45m Biology Lessons - Lesson 67</div></div>                                                                                     |  |
| <div><div>Exam Week</div></div>                                                                                                                                                    |  |
| <div><div>→ EXAMS</div><div>Answer question(s) related to the course. Complete about 3 questions per day or as directed by your teacher.</div></div>                               |  |
| <div><div>WEEK 12</div><div><div><div></div></div> 45m Biology Lessons - Lesson 68</div></div>                                                                                     |  |
| <div><div>Exam Week</div></div>                                                                                                                                                    |  |
| <div><div>→ EXAMS</div><div>Answer question(s) related to the course.</div></div>                                                                                                  |  |
| <div><div>WEEK 12</div><div><div><div></div></div> 45m Biology Lessons - Lesson 69</div></div>                                                                                     |  |
| <div><div>Exam Week</div></div>                                                                                                                                                    |  |
| <div><div>→ EXAMS</div><div>Answer question(s) related to the course.</div></div>                                                                                                  |  |
| <div><div>WEEK 12</div><div><div><div></div></div> 45m Biology Lessons - Lesson 70</div></div>                                                                                     |  |
| <div><div>Exam Week</div></div>                                                                                                                                                    |  |
| <div><div>→ EXAMS</div><div>Answer question(s) related to the course.</div></div>                                                                                                  |  |
| <div><div>WEEK 12</div><div><div><div></div></div> 45m Biology Lessons - Lesson 71</div></div>                                                                                     |  |
| <div><div>Exam Week</div></div>                                                                                                                                                    |  |
| <div><div>→ EXAMS</div><div>Answer question(s) related to the course.</div></div>                                                                                                  |  |
| <div><div>WEEK 12</div><div><div><div></div></div> 60m Biology Lessons - Lesson 72</div></div>                                                                                     |  |
| <div><div>Exam Week</div></div>                                                                                                                                                    |  |
| <div><div>→ EXAMS</div><div>Answer question(s) related to the course.</div></div>                                                                                                  |  |

# Science: Biology

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## Term 2

### WEEK 1 45m Biology Lessons - Lesson 73

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Read the Introduction to Ch.6 and the Objectives for Section 6.1. Then read Section 6.1.1 and part of 6.1.2 (at least the first 2 paragraphs and study Figure 6.5). Note important terms and their definitions, copy any helpful diagrams, and record any interesting or important points. Remember to do this as you continue throughout the course.

#### • PLAN WEEKLY

- ☐ nature walk - record observations, save a leaf for later in the term
- ☐ science free read

### WEEK 1 45m Biology Lessons - Lesson 74

Materials: General Biology and Coloring Book

#### → READ, NARRATE, & DISCUSS

Finish section 6.1.2. Then color plate 66.

Note that there is much history to enjoy in plates 68-72, though these are not required!

#### ★ STUDENT REMINDER

Remember to break any overwhelming processes or figures into smaller parts or steps and focus on getting at the ideas rather than memorizing.

### WEEK 1 45m Biology Lessons - Lesson 75

Materials: General Biology and Coloring Book

#### → READ, NARRATE, & DISCUSS

Section 6.1.3-6.2, referring to the Objectives for Section 6.2 at the beginning of the chapter, before beginning that section. Continue coloring as time permits.

### WEEK 1 45m Biology Lessons - Lesson 76

Materials: General Biology

#### → PRACTICE

Begin working through the review exercises at the end of Ch.6.

#### ★ STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

### WEEK 1 45m Biology Lessons - Lesson 77

Materials: General Biology

#### → PRACTICE

Continue working through the review exercises at the end of Ch.6.

### WEEK 1 60m Biology Lessons - Lesson 78

*Observing Meiosis*

Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Complete Lab 9.

### WEEK 2 45m Biology Lessons - Lesson 79

Materials: General Biology and Coloring Book

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 6.3 at the beginning of the chapter. Then read Section 6.3.1.

Color plates 61, 64, and 73-75 as time permits.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

# Science: Biology

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## Term 2

|                                                                                                                                                                                                                        |                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>WEEK 2</b> 45m Biology Lessons - Lesson 80</p> <p>Materials: General Biology</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Section 6.3.2. Work review exercises and/or continue coloring as time permits.</p> | <p>★ <b>STUDENT REMINDER</b><br/>Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.</p> |
| <p><b>WEEK 2</b> 45m Biology Lessons - Lesson 81</p> <p>Materials: General Biology</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Section 6.3.3. Work review exercises and/or continue coloring as time permits.</p> |                                                                                                                                                       |
| <p><b>WEEK 2</b> 45m Biology Lessons - Lesson 82</p> <p>Materials: General Biology</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Section 6.3.4. Work review exercises as time permits.</p>                          | <p>★ <b>STUDENT REMINDER</b><br/>Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!</p>    |
| <p><b>WEEK 2</b> 45m Biology Lessons - Lesson 83</p> <p>Materials: General Biology</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Section 6.3.5. Work review exercises as time permits.</p>                          |                                                                                                                                                       |
| <p><b>WEEK 2</b> 60m Biology Lessons - Lesson 84<br/><i>Investigating Cat Coat Phenotypes</i></p> <p>Materials: High School Biology Lab Book and materials listed within</p> <p>→ <b>LAB DAY</b><br/>Begin Lab 10.</p> | <p>★ <b>STUDENT/TEACHER NOTE</b><br/>Lab 10 will be completed over three lessons. (Lesson 84, 89, and 90)</p>                                         |
| <p><b>WEEK 3</b> 45m Biology Lessons - Lesson 85</p> <p>Materials: General Biology</p> <p>→ <b>PRACTICE</b><br/>Continue working through the review exercises at the end of Ch.6.</p>                                  | <p>● <b>PLAN WEEKLY</b><br/>nature walk - record observations<br/>science free read</p>                                                               |
| <p><b>WEEK 3</b> 45m Biology Lessons - Lesson 86</p> <p>Materials: General Biology</p> <p>→ <b>PRACTICE</b><br/>Continue working through the review exercises at the end of Ch.6.</p>                                  |                                                                                                                                                       |
| <p><b>WEEK 3</b> 45m Biology Lessons - Lesson 87</p> <p>Materials: General Biology</p> <p>→ <b>PRACTICE</b><br/>Continue working through the review exercises at the end of Ch.6.</p>                                  |                                                                                                                                                       |



## Term 2

### WEEK 3 45m Biology Lessons - Lesson 88

Materials: General Biology

#### → REVIEW

Take some time to review Ch.6 and think about how it connects to previous chapters in the book. Look back over your notes and the chapter objectives. Review anything difficult to recall.

### WEEK 3 45m Biology Lessons - Lesson 89

*Investigating Cat Coat Phenotypes*

Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Continue Lab 10.

### WEEK 3 60m Biology Lessons - Lesson 90

*Investigating Cat Coat Phenotypes*

Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Complete Lab 10.

### WEEK 4 45m Biology Lessons - Lesson 91

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Read the Introduction to Ch.7 and the Objectives for Section 7.1 at the beginning of the chapter. Then read Section 7.1.

Work review exercises as time permits.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 4 45m Biology Lessons - Lesson 92

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 7.2 at the beginning of the chapter. Then read Section 7.2.1-7.2.3.

#### ★ STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

### WEEK 4 45m Biology Lessons - Lesson 93

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Section 7.2.4-7.2.5. Work review exercises as time permits.

### WEEK 4 45m Biology Lessons - Lesson 94

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 7.3 at the beginning of the chapter. Then read Section 7.3.1-7.3.2. Work review exercises as time permits.

#### ★ STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

# Science: Biology

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## Term 2

### WEEK 4 45m Biology Lessons - Lesson 95

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Section 7.3.3-7.3.5. Work review exercises as time permits.

### WEEK 4 60m Biology Lessons - Lesson 96

*Discovering Bacteria Among Us*

Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Begin Lab 11.

#### ★ STUDENT/TEACHER NOTE

Lab 11 will be completed over two lessons. (Lessons 96 and 102)

### WEEK 5 45m Biology Lessons - Lesson 97

Materials: General Biology

#### → REVIEW

Continue working through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments.

#### ● PLAN WEEKLY

- nature walk - record observations
- science free read

### WEEK 5 45m Biology Lessons - Lesson 98

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 7.4 at the beginning of the chapter. Then read Section 7.4.1-7.4.2.

#### ● STUDENT TIP

Be sure to look at the Citizen Science project that begins next week to be sure you have the necessary materials!

### WEEK 5 45m Biology Lessons - Lesson 99

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Section 7.4.3-7.4.4. Work review exercises as time permits.

#### ★ STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

### WEEK 5 45m Biology Lessons - Lesson 100

Materials: General Biology

PREP: Read Teacher Tip

#### → REVIEW

Continue working through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments.

### WEEK 5 45m Biology Lessons - Lesson 101

Materials: General Biology

#### → REVIEW

Continue working through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments.



## Term 2

### WEEK 5 ☐ 60m Biology Lessons - Lesson 102

*Discovering Bacteria Among Us*

☐ Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Complete Lab 11.

### WEEK 6 ☐ 45m Biology Lessons - Lesson 103

☐ Materials: Website link and notebook/computer to collect data

#### → SET UP

Use today to set up the Sourdough for Science Project and become familiar with the data collection. You will need to collect data for 14 consecutive days. Data collection should only take a few minutes, so you can do this at the start of your lesson time.

∞ Website Link: Sourdough for Citizen Science

#### • PLAN WEEKLY

☐ nature walk - record observations

☐ science free read

#### ★ TEACHER TIP

You may want to bookmark the website link so it can be referred to easily. It is referenced, but not linked in Lessons 103-107, 109-113, and 115-120.

### WEEK 6 ☐ 45m Biology Lessons - Lesson 104

☐ Materials: Website link, notebook/computer to collect data, and General Biology

#### → DATA COLLECTION

Spend a few minutes collecting data according to the instructions on the website.

#### → READ, NARRATE, & DISCUSS

Read the Introduction to Ch.8 and the Objectives for Section 8.1 at the beginning of the chapter. Then read to about half of section 8.1.3.

### WEEK 6 ☐ 45m Biology Lessons - Lesson 105

☐ Materials: Website link, notebook/computer to collect data, and General Biology

#### → DATA COLLECTION

Spend a few minutes collecting data according to the instructions on the website.

#### → READ, NARRATE, & DISCUSS

Finish Section 8.1. Begin working on exercises as time permits.

### WEEK 6 ☐ 45m Biology Lessons - Lesson 106

☐ Materials: Website link, notebook/computer to collect data, and General Biology

#### → DATA COLLECTION

Spend a few minutes collecting data according to the instructions on the website.

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 8.2-8.3 at the beginning of the chapter. Then read Section 8.2-8.3.1. Continue review exercises as time permits.

#### ★ STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.



## Term 2

### WEEK 6 45m Biology Lessons - Lesson 107

Materials: Website link, notebook/computer to collect data, and General Biology

#### → DATA COLLECTION

Spend a few minutes collecting data according to the instructions on the website.

#### → READ, NARRATE, & DISCUSS

Sections 8.3.2-8.3.3. Continue review exercises as time permits.

#### ★ STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

### WEEK 6 60m Biology Lessons - Lesson 108

*Using a Dichotomous Key*

Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Complete Lab 12.

### WEEK 7 45m Biology Lessons - Lesson 109

Materials: Website link, notebook/computer to collect data, and General Biology

#### → DATA COLLECTION

Spend a few minutes collecting data according to the instructions on the website.

∞ Website Link: Sourdough for Citizen Science

#### → READ, NARRATE, & DISCUSS

Historical background "Guided Reading" that begins on p.20 of Sourdough for Science Teacher's Guide.

∞ Document Link: Sourdough Teacher Guide

Continue Ch.8 review exercises as time permits.

#### ● PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 7 45m Biology Lessons - Lesson 110

Materials: Website link, notebook/computer to collect data, and General Biology

#### → DATA COLLECTION

Spend a few minutes collecting data according to the instructions on the website.

#### → READ, NARRATE, & DISCUSS

Section 8.3.4. Continue Ch.8 review exercises as time permits.

### WEEK 7 45m Biology Lessons - Lesson 111

Materials: Website link, notebook/computer to collect data, and General Biology

#### → DATA COLLECTION

Spend a few minutes collecting data according to the instructions on the website.

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 8.4 at the beginning of the chapter. Then read to the end of section 8.4.2.

#### ★ STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.



## Term 2

### WEEK 7 45m Biology Lessons - Lesson 112

Materials: Website link, notebook/computer to collect data, and General Biology

#### → DATA COLLECTION

Spend a few minutes collecting data according to the instructions on the website.

#### → READ, NARRATE, & DISCUSS

Section 8.4.3. Continue review exercises as time permits.

### WEEK 7 45m Biology Lessons - Lesson 113

Materials: Website link, notebook/computer to collect data, and General Biology

#### → DATA COLLECTION

Spend a few minutes collecting data according to the instructions on the website.

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 8.5 at the beginning of the chapter. Then read Section 8.5. Continue review exercises as time permits.

#### ★ STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

### WEEK 7 60m Biology Lessons - Lesson 114

*Making Sense of Plant Histology*

Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Complete Lab 13.

### WEEK 8 45m Biology Lessons - Lesson 115

Materials: Website link, notebook/computer to collect data, and General Biology

#### → DATA COLLECTION

Spend a few minutes collecting data according to the instructions on the website.

∞ Website Link: Sourdough for Citizen Science

#### → REVIEW

Continue working through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 8 45m Biology Lessons - Lesson 116

Materials: Website link, notebook/computer to collect data, and General Biology

#### → DATA COLLECTION

Spend a few minutes collecting data according to the instructions on the website.

#### → REVIEW

Continue working through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments.



## Term 2

### WEEK 8 45m Biology Lessons - Lesson 117

Materials: Website link and notebook or computer

#### → DATA ANALYSIS

Plot your data versus time as shown on the project homepage. You can do this on the graph paper in your notebook or on a spreadsheet. Upload your data to the project website.

∞ Website Link: Sourdough for Citizen Science

### WEEK 8 45m Biology Lessons - Lesson 118

Materials: Website link

#### → POETRY

Complete the activity entitled "Wonderbread" beginning on p.25 of the Teacher's Guide.

∞ Document Link: Sourdough Teacher Guide

Then use the remaining time to continue working through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments.

### WEEK 8 45m Biology Lessons - Lesson 119

Materials: Website link and notebook/computer to compose

#### → REPORT WRITING

Prepare a composition to report on your citizen science project. This could be an expository essay, a PowerPoint or poster presentation, a Canva or watercolor graphic display, or any other form of expression that allows you to effectively communicate what you have learned. Effective communication of complex ideas is a major challenge in science, so think about how you communicate best and plan your composition thoughtfully. Be sure to include your results.

#### ★ STUDENT/TEACHER NOTE

In the next lesson, Lesson 120, students will finish their citizen science project report started in today's lesson.

### WEEK 8 60m Biology Lessons - Lesson 120

Materials: Website link and notebook/computer to compose

#### → COMPOSE

Complete your report on this project using whatever form of composition you chose.

### WEEK 9 45m Biology Lessons - Lesson 121

Materials: General Biology

#### → REVIEW

Take some time to do a cumulative review of what you have learned so far. Look back over your notes and the chapter objectives. Review anything difficult to recall.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 9 45m Biology Lessons - Lesson 122

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Read the Introduction to Ch.9 and the Objectives for Section 9.1 at the beginning of the chapter. Then read to the end of section 9.1.2.



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### WEEK 9 45m Biology Lessons - Lesson 123

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Section 9.1.3. Then begin working on exercises for Section 9.1.

#### ★ STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

### WEEK 9 45m Biology Lessons - Lesson 124

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 9.2 at the beginning of the chapter. Then read to the end of section 9.2.2. Continue exercises as time permits.

### WEEK 9 45m Biology Lessons - Lesson 125

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Sections 9.2.3. Then, continue working on review exercises as time permits.

#### ★ STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

### WEEK 9 60m Biology Lessons - Lesson 126

*Dissecting an Earthworm*

Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Complete Lab 14.

### WEEK 10 45m Biology Lessons - Lesson 127

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Sections 9.2.4-9.2.5. Then, continue working on review exercises as time permits.

#### ● PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 10 45m Biology Lessons - Lesson 128

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Sections 9.2.6. Then, continue working on review exercises as time permits.

### WEEK 10 45m Biology Lessons - Lesson 129

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Sections 9.2.7-9.2.8. Then, continue working on review exercises as time permits.

#### ★ STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

### WEEK 10 45m Biology Lessons - Lesson 130

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 9.3 at the beginning of the chapter.



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Then read Section 9.3.1-9.3.3. If you don't finish all of 9.3.3, you will have more time in the next lesson. Continue exercises as time permits.

### WEEK 10 45m Biology Lessons - Lesson 131

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Sections 9.3.3-9.3.4. Then, continue working on review exercises as time permits.

### WEEK 10 60m Biology Lessons - Lesson 132

*Dissecting a Crayfish*

Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Complete Lab 15.

### WEEK 11 45m Biology Lessons - Lesson 133

Materials: General Biology

#### → REVIEW & CATCH-UP

Work through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 11 45m Biology Lessons - Lesson 134

Materials: General Biology

PREP: Read Teacher Tip

#### → REVIEW & CATCH-UP

Work through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments.

### WEEK 11 45m Biology Lessons - Lesson 135

Materials: General Biology

#### → REVIEW & CATCH-UP

Work through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments.

### WEEK 11 45m Biology Lessons - Lesson 136

Materials: General Biology

#### → REVIEW & CATCH-UP

Work through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments.

### WEEK 11 45m Biology Lessons - Lesson 137

Materials: General Biology

#### → REVIEW & CATCH-UP

Work through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments.



## Term 2

### WEEK 11 60m Biology Lessons - Lesson 138

Materials: General Biology

#### → REVIEW & CATCH-UP

Work through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments.

### WEEK 12 45m Biology Lessons - Lesson 139

*Exam Week*

#### → EXAMS

Answer question(s) related to the course. Complete about 3 questions per day or as directed by your teacher.

### WEEK 12 45m Biology Lessons - Lesson 140

*Exam Week*

#### → EXAMS

Answer question(s) related to the course.

### WEEK 12 45m Biology Lessons - Lesson 141

*Exam Week*

#### → EXAMS

Answer question(s) related to the course.

### WEEK 12 45m Biology Lessons - Lesson 142

*Exam Week*

#### → EXAMS

Answer question(s) related to the course.

### WEEK 12 45m Biology Lessons - Lesson 143

*Exam Week*

#### → EXAMS

Answer question(s) related to the course.

### WEEK 12 60m Biology Lessons - Lesson 144

*Exam Week*

#### → EXAMS

Answer question(s) related to the course.



## Term 3

### WEEK 1 45m Biology Lessons - Lesson 145

Materials: Current Event link

#### → INTRO

Term 2 of General Biology brought us to the study of the Kingdom Animalia, which includes humans. Just as we have done with other living organisms, we will study the anatomy (the structures) and the physiology (the function). The relationship between structure and function is important in Biology. Can you explain why? Read and discuss the current event article below.

#### → READ, NARRATE, & DISCUSS

Read the following article from Science News Explores, "Breeding has Given Dogs Distinct Brain Shapes."  
∞ Article Link: Breeding

Review course objectives from previous chapters with any time remaining.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 1 45m Biology Lessons - Lesson 146

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Read the Introduction to Ch.10 and the Objectives for Section 10.1 at the beginning of the chapter. Then read to the end of section 10.1.3.

### WEEK 1 45m Biology Lessons - Lesson 147

Materials: Coloring Book

#### → COLOR

Begin coloring plates 98, 100, and 101

### WEEK 1 45m Biology Lessons - Lesson 148

Materials: General Biology, Microscope, and Prepared Slide Set

#### → READ, NARRATE, & DISCUSS

Sections 10.1.4-10.1.5.

Take a brief look at the different types of muscle cells in your prepared slide set (#14, 16, and 20). Then continue exercises as time permits.

#### ★ STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

### WEEK 1 45m Biology Lessons - Lesson 149

*Dissecting a Frog*

Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Begin Lab 16.

#### ★ STUDENT/TEACHER NOTE

Lab 16 will be completed over four lessons. Two lessons in Week 1 and two lessons in Week 2.

### WEEK 1 60m Biology Lessons - Lesson 150

*Dissecting a Frog*

Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Continue Lab 16.



## Term 3

### WEEK 2 45m Biology Lessons - Lesson 151

Materials: General Biology, Microscope, and Prepared Slide Set

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 10.2 at the beginning of the chapter. Then read Section 10.2.

Take a brief look at the motor nerve from your prepared slide set (#17). Continue exercises as time permits.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 2 45m Biology Lessons - Lesson 152

Materials: General Biology and Coloring Book

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 10.3 at the beginning of the chapter. Then read to the end of section 10.3.2.

Then, begin coloring plates 102, 105, and 110.

#### ★ STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

### WEEK 2 45m Biology Lessons - Lesson 153

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Section 10.3.3-10.4.3, referring to the Objectives for Section 10.4 at the beginning of the chapter before beginning that section. Continue review exercises as time permits.

### WEEK 2 45m Biology Lessons - Lesson 154

Materials: General Biology and Coloring Book

#### → REVIEW

Continue coloring unfinished plates and/or plate 107. Then, continue review exercises for Ch.10.

### WEEK 2 45m Biology Lessons - Lesson 155

*Dissecting a Frog*

Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Continue Lab 16.

### WEEK 2 60m Biology Lessons - Lesson 156

*Dissecting a Frog*

Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Complete Lab 16.

### WEEK 3 45m Biology Lessons - Lesson 157

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Section 10.4.4-10.5.3, referring to the Objectives for Section 10.5 at the

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read



## Term 3

beginning of the chapter before beginning that section. Continue review exercises as time permits.

### WEEK 3 45m Biology Lessons - Lesson 158

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 10.6 at the beginning of the chapter. Then read Section 10.6. Continue review exercises as time permits.

### WEEK 3 45m Biology Lessons - Lesson 159

Materials: General Biology and Coloring Book

#### → REVIEW

Continue coloring unfinished plates and/or plates 106 and 109. Then, continue review exercises for Ch.10.

#### ★ STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

### WEEK 3 45m Biology Lessons - Lesson 160

Materials: General Biology and Coloring Book

#### → REVIEW

Continue coloring unfinished plates. Then, continue review exercises for Ch.10.

### WEEK 3 45m Biology Lessons - Lesson 161

Materials: internet or library access

#### → INTRO

You will spend several days researching a human disease or medical condition. After your research, you will compose an essay or presentation to teach others what you have learned.

#### → RESEARCH

Begin your research today using any combination of literature, the internet, or professional interviews.

- Do scientists know the cause(s) of the disease or condition? If so, what is it? If not, what theories do they have?
- How does the disease or condition affect the body?
- What treatments are available?
- What can people do to help prevent any negative effects?

#### ★ STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

### WEEK 3 60m Biology Lessons - Lesson 162

Materials: internet or library access

#### → RESEARCH

Continue your research.

### WEEK 4 45m Biology Lessons - Lesson 163

Materials: internet or library access

#### → RESEARCH

Continue your research.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read



## Term 3

### WEEK 4 ☐ 45m Biology Lessons - Lesson 164

☐ Materials: notebook or computer to compose

#### → REPORT WRITING

Prepare a composition to report on your medical research. This could be an expository essay, PowerPoint or poster presentation, a graphic display using Canva or watercolor, or any other form of expression that allows you to communicate what you have learned effectively. Effective communication of complex ideas is a major challenge in science, so think about how you communicate best and plan your composition thoughtfully!

### WEEK 4 ☐ 45m Biology Lessons - Lesson 165

☐ Materials: notebook or computer to compose

#### → REPORT WRITING

Complete your report on your medical research.

### WEEK 4 ☐ 45m Biology Lessons - Lesson 166

☐ Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Read the Introduction to Ch.11 and the Objectives for Sections 11.1-11.2 at the beginning of the chapter. Then read Sections 11.1-11.2.1. Begin review exercises as time permits.

#### ★ STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

### WEEK 4 ☐ 45m Biology Lessons - Lesson 167

☐ Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Sections 11.2.2. Work on exercises for Ch.11.

### WEEK 4 ☐ 60m Biology Lessons - Lesson 168

*Considering Biomes*

☐ Materials: General Biology, Article link, and the NatGeo atlas

#### → LAB DAY

Compare the map from the National Geographic link with the one in your text on p.330.

∞ Website Link: Biomes

Understanding that the definition and boundaries of the biome are somewhat flexible, characterize the biome where you live, school, or work. You need to consider the climate and geography, and include at least 3 common plants and animals representing your biome.

### WEEK 5 ☐ 45m Biology Lessons - Lesson 169

☐ Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 11.3 at the beginning of the chapter. Then read Sections 11.3.1-11.3.4. Begin review exercises as time permits.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read



## Term 3

### WEEK 5 45m Biology Lessons - Lesson 170

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Section 11.3.5. Then, continue working on the review exercises as time permits.

#### ★ STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

### WEEK 5 45m Biology Lessons - Lesson 171

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 11.4 at the beginning of the chapter. Then read Sections 11.4-11.4.1. Continue review exercises as time permits.

### WEEK 5 45m Biology Lessons - Lesson 172

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Sections 11.4.2. Then, continue working on the review exercises as time permits.

### WEEK 5 45m Biology Lessons - Lesson 173

Materials: internet or library access

#### → INTRO

Building on your biome characterization from last week, identify one threatened or endangered species in your biome to learn more about. You could do this via the Internet or by consulting a local park ranger or conservator.

#### → RESEARCH

- Your species lives in what habitat(s)?
- What does your species eat?
- What preys on your species?
- What has caused your species to be threatened or endangered?
- Plan to visit the species at a conservatory or museum and/or its natural habitat during your next outdoor visit.

### WEEK 5 60m Biology Lessons - Lesson 174

Materials: internet or library access

#### → RESEARCH

- What is being done (or considered) to support your threatened/endangered species?
- What are the pros/cons of these strategies for everyone in the ecosystem (i.e., plants and animals, including humans)?
- Consider local resources and what citizens can do right now to do the most good and the least harm for all concerned.

### WEEK 6 45m Biology Lessons - Lesson 175

Materials: internet or library access, and The Fool and the Heretic/The Spirit of the Rainforest (science free read)

#### → RESEARCH OR FREE READ

If you are not finished with your research, do so today. Otherwise, read

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read



## Term 3

your science free read.

### WEEK 6 45m Biology Lessons - Lesson 176

Materials: General Biology

PREP: Read Teacher Tip

#### → READ, NARRATE, & DISCUSS

Sections 11.4.3. Then, continue working on the review exercises as time permits.

### WEEK 6 45m Biology Lessons - Lesson 177

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Sections 11.4.4. Then, continue working on the review exercises as time permits.

#### ★ STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

### WEEK 6 45m Biology Lessons - Lesson 178

Materials: General Biology

#### → REVIEW

Continue working through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments.

### WEEK 6 45m Biology Lessons - Lesson 179

Materials: notebook or computer to compose

#### → REPORT WRITING

Prepare a composition to report on your ecological research. This could be an expository essay, a PowerPoint for a poster presentation, a graphic display using Canva or watercolor, or any other form of expression that allows you to communicate what you have learned effectively. Effective communication of complex ideas is a major challenge in science, so think about how you communicate best and plan your composition thoughtfully!

### WEEK 6 60m Biology Lessons - Lesson 180

Materials: notebook or computer to compose

PREP: Read Student/Teacher Tip

#### → REPORT WRITING

Complete your report on your ecological research.

#### ★ STUDENT/TEACHER TIP

Note that students will repeat their BioBlitz next week. The 2 days of outdoor work can be moved to any day of the week to accommodate the weather.

### WEEK 7 45m Biology Lessons - Lesson 181

Materials: General Biology and/or The Fool and the Heretic/The Spirit of the Rainforest (science free read)

#### → REVIEW

Continue working through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments.

Continue your science free read.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read



## Term 3

### WEEK 7 45m Biology Lessons - Lesson 182

Materials: article link

#### → INTRO

You will repeat your BioBlitz this week to compare what you find in different seasons. For reference, the NatGeo BioBlitz article is in the links.  
∞ Article Link: BioBlitz

#### → FIELDWORK

Your BioBlitz will last for 1 day, so choose your area carefully based on how many scientists will keep inventory and what you plan to record. You may inventory all species or focus on only one type of object (e.g., plants, insects, birds, etc.). You will only briefly observe, identify, and document each object in your lab notebook - you will not journal on individuals as you do in nature study. Today, decide what you will inventory and plan how you will collect your BioBlitz data. Collect any special materials or equipment that you might need. Then, begin by writing an introductory narration in your notebook and getting an overview of the area. You might want to sketch a landscape or map the area, noting any landmarks.

### WEEK 7 45m Biology Lessons - Lesson 183

Materials: graph paper or spreadsheet

#### → FIELDWORK

Collect your BioBlitz data.

### WEEK 7 45m Biology Lessons - Lesson 184

Materials: General Biology

#### → REVIEW

Continue working through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments.

### WEEK 7 45m Biology Lessons - Lesson 185

Materials: General Biology and The Fool and the Heretic/The Spirit of the Rainforest (science free read)

#### → REVIEW OR FREE READ

Continue working through review exercises, review notes/narrations, and course objectives, and/or complete any unfinished assignments. Or, read your science free read.

### WEEK 7 60m Biology Lessons - Lesson 186

Materials: graph paper or spreadsheet

#### → DATA ANALYSIS

Analyze your BioBlitz data as you did in Term 1 of Biology. Compare your results. Do your observations raise any questions or support any hypotheses?

### WEEK 8 45m Biology Lessons - Lesson 187

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Read the Introduction to Ch.12 and the Objectives for Section 12.1 at the beginning of the chapter. Then read Section 12.1.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read



## Term 3

### WEEK 8 45m Biology Lessons - Lesson 188

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 12.2 at the beginning of the chapter. Then read Section 12.2. Begin review exercises as time permits.

#### ★ STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

### WEEK 8 45m Biology Lessons - Lesson 189

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 12.3 at the beginning of the chapter. Then read Section 12.3. Work review exercises as time permits.

### WEEK 8 45m Biology Lessons - Lesson 190

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Refer to the Objectives for Section 12.4 at the beginning of the chapter. Then read Section 12.4. Continue review exercises as time permits.

#### ★ STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

### WEEK 8 45m Biology Lessons - Lesson 191

*Simulating Natural Selection*

Materials: High School Biology Lab Book and materials listed within

#### → LAB DAY

Complete Lab 17. If you need additional time, this lab may continue into lesson 192.

### WEEK 8 60m Biology Lessons - Lesson 192

*Simulating Natural Selection*

Materials: High School Biology Lab Book, materials listed within Lab Book, and The Fool and the Heretic/The Spirit of the Rainforest (science free read)

#### → LAB DAY OR FREE READ

Continue/complete Lab 17. Then, read your science free read.

### WEEK 9 45m Biology Lessons - Lesson 193

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Ch.13 Preface (this is after the Objectives) and Section 13.1.1. Then refer to the Objectives for Section 13.1 at the beginning of the chapter before reading Section 13.1.2.

#### ● PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 9 45m Biology Lessons - Lesson 194

Materials: General Biology

#### → READ, NARRATE, & DISCUSS

Read to the end of Section 13.1. Begin review exercises as time permits.



## Term 3

|                                                                                                                                                                                                                                                                                                             |                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>WEEK 9</b> 45m Biology Lessons - Lesson 195</p> <p>Materials: General Biology</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Refer to the Objectives for Section 13.2. Then read to the end of section 13.2.1. Continue review exercises as time permits.</p>                                       | <p>★ <b>STUDENT REMINDER</b><br/>Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!</p>    |
| <p><b>WEEK 9</b> 45m Biology Lessons - Lesson 196</p> <p>Materials: General Biology</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Sections 13.2.2-13.2.3. Continue review exercises as time permits.</p>                                                                                                 |                                                                                                                                                       |
| <p><b>WEEK 9</b> 45m Biology Lessons - Lesson 197</p> <p>Materials: General Biology</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Sections 13.2.4-13.2.5. Continue review exercises as time permits.</p>                                                                                                 | <p>★ <b>STUDENT REMINDER</b><br/>Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.</p> |
| <p><b>WEEK 9</b> 60m Biology Lessons - Lesson 198</p> <p>Materials: General Biology, Coloring Book, The Fool and the Heretic/The Spirit of the Rainforest (science free read)</p> <p>→ <b>PRACTICE, COLOR, OR FREE READ</b><br/>Continue review exercises, Plates 89-90, and/or your science free read.</p> |                                                                                                                                                       |
| <p><b>WEEK 10</b> 45m Biology Lessons - Lesson 199</p> <p>Materials: General Biology, Coloring Book, The Fool and the Heretic/The Spirit of the Rainforest (science free read)</p> <p>→ <b>PRACTICE, COLOR, OR FREE READ</b><br/>Continue review exercises, Plates 89-90, or your science free read.</p>    | <p>● <b>PLAN WEEKLY</b></p> <ul style="list-style-type: none"> <li>nature walk - record observations</li> <li>science free read</li> </ul>            |
| <p><b>WEEK 10</b> 45m Biology Lessons - Lesson 200</p> <p>Materials: General Biology, Coloring Book, The Fool and the Heretic/The Spirit of the Rainforest (science free read)</p> <p>→ <b>PRACTICE, COLOR, OR FREE READ</b><br/>Continue review exercises, Plates 89-90, or your science free read.</p>    |                                                                                                                                                       |
| <p><b>WEEK 10</b> 45m Biology Lessons - Lesson 201</p> <p>Materials: General Biology, Coloring Book, The Fool and the Heretic/The Spirit of the Rainforest (science free read)</p> <p>→ <b>PRACTICE, COLOR, OR FREE READ</b><br/>Continue review exercises, Plates 89-90, or your science free read.</p>    |                                                                                                                                                       |
| <p><b>WEEK 10</b> 45m Biology Lessons - Lesson 202</p> <p>Materials: General Biology, Coloring Book, The Fool and the Heretic/The Spirit of the Rainforest (science free read)</p> <p>→ <b>PRACTICE, COLOR, OR FREE READ</b><br/>Continue review exercises, Plates 89-90, or your science free read.</p>    |                                                                                                                                                       |



## Term 3

### WEEK 10 ☐ 45m Biology Lessons - Lesson 203

☐ Materials: General Biology

#### → REVIEW

Begin your cumulative review for the year. Review exercises, notes/narrations, course objectives, coloring pages, labs, etc. Re-read anything that you may have forgotten and consider how all the work for the course fits together.

### WEEK 10 ☐ 60m Biology Lessons - Lesson 204

☐ Materials: General Biology

#### → REVIEW

Continue your cumulative review for the year. Review exercises, notes/narrations, course objectives, coloring pages, labs, etc. Re-read anything that you may have forgotten and consider how all the work for the course fits together.

### WEEK 11 ☐ 45m Biology Lessons - Lesson 205

☐ Materials: General Biology or The Fool and the Heretic/The Spirit of the Rainforest (science free read)

#### → REVIEW & CATCH-UP OR FREE READ

Complete any unfinished assignments and review for the entire year. Or, read your science free read.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 11 ☐ 45m Biology Lessons - Lesson 206

☐ Materials: General Biology or The Fool and the Heretic/The Spirit of the Rainforest (science free read)

PREP: Read Teacher Tip

#### → REVIEW & CATCH-UP OR FREE READ

Complete any unfinished assignments and review for the entire year. Or, read your science free read.

### WEEK 11 ☐ 45m Biology Lessons - Lesson 207

☐ Materials: General Biology or The Fool and the Heretic/The Spirit of the Rainforest (science free read)

#### → REVIEW & CATCH-UP OR FREE READ

Complete any unfinished assignments and review for the entire year. Or, read your science free read.

### WEEK 11 ☐ 45m Biology Lessons - Lesson 208

☐ Materials: General Biology or The Fool and the Heretic/The Spirit of the Rainforest (science free read)

#### → REVIEW & CATCH-UP OR FREE READ

Complete any unfinished assignments and review for the entire year. Or, read your science free read.



## Term 3

### WEEK 11 45m Biology Lessons - Lesson 209

Materials: General Biology or The Fool and the Heretic/The Spirit of the Rainforest (science free read)

#### → REVIEW & CATCH-UP OR FREE READ

Complete any unfinished assignments and review for the entire year. Or, read your science free read.

### WEEK 11 60m Biology Lessons - Lesson 210

Materials: General Biology or The Fool and the Heretic/The Spirit of the Rainforest (science free read)

#### → REVIEW & CATCH-UP OR FREE READ

Complete any unfinished assignments and review for the entire year. Or, read your science free read.

### WEEK 12 45m Biology Lessons - Lesson 211

*Exam Week*

#### → EXAMS

Answer question(s) related to the course. Complete about 3 questions per day or as directed by your teacher.

### WEEK 12 45m Biology Lessons - Lesson 212

*Exam Week*

#### → EXAMS

Answer question(s) related to the course.

### WEEK 12 45m Biology Lessons - Lesson 213

*Exam Week*

#### → EXAMS

Answer question(s) related to the course.

### WEEK 12 45m Biology Lessons - Lesson 214

*Exam Week*

#### → EXAMS

Answer question(s) related to the course.

### WEEK 12 45m Biology Lessons - Lesson 215

*Exam Week*

#### → EXAMS

Answer question(s) related to the course.

### WEEK 12 60m Biology Lessons - Lesson 216

*Exam Week*

#### → EXAMS

Answer question(s) related to the course.

# Science: Biology

## Examination

### Term 1

#### Biology Lessons

- General Biology--choose 2 exercises from the following:
  - Ch.1 Exercises-#3, #16, Ch.2 Exercises-#5, #16
- General Biology--Choose 2 exercises from the following:  
(You may substitute the previous day's questions as appropriate based on the material covered)
  - Ch.2 Exercise-#28, Ch.3 Exercises-#7, #22, or #30
- General Biology--Choose 2 exercises from the following:  
(You may substitute previous days' questions as appropriate based on the material covered)
  - Ch.4 Exercises-#8, #36, Ch.5 Exercises-#22, #39
- Explain one of your laboratory exercises from this term and what you learned from it. Describe any questions this made you think of.
- Explain an example of biology in the world around you that you noticed this term. This might be from your project work, current events, nature walks, or your free read. Describe any questions this made you think of.

### Term 2

#### Biology Lessons

- General Biology--Choose 2 exercises from the following:  
(You may substitute the previous term's questions as appropriate based on the material covered.)
  - Ch.6 Exercises-#4, #28, #33, or #35
- General Biology--Choose 2 exercises from the following:  
(You may substitute questions from the previous day or terms as appropriate, based on material covered)
  - Ch.7 Exercises-#4, #12, #30, or #34
- General Biology--Choose 2 exercises from the following:  
(You may substitute questions from the previous days or terms as appropriate, based on material covered)
  - Ch.8 Exercises-#17, #29, Ch.9 Exercises-#8, #57
- Explain one of your laboratory exercises from this term and what you learned from it. Describe any questions this made you think of.
- Explain an example of biology in the world around you that you noticed this term. This might be from your project work, current events, nature walks, or your free read. Describe any questions this made you think of.

### Term 3

#### Biology Lessons

- General Biology--Choose 2 exercises from the following:  
(You may substitute questions from the previous terms as appropriate, based on material covered.)
  - Ch.10 Exercises-#5, #19, #23, or #35
- General Biology--Choose 2 exercises from the following:  
(You may substitute questions from the previous day or terms as appropriate, based on material covered)
  - Ch.11 Exercises-#4, #10, #32, or #47
- General Biology--Choose 2 exercises from the following:  
(You may substitute questions from the previous days or terms as appropriate, based on material covered)
  - Ch.12 Exercises-#6, Ch.13 #5, #6, or #12
- Explain one of your laboratory exercises from this term and what you learned from it. Describe any questions this made you think of.
- Explain an example of biology in the world around you that you noticed this term. This might be from your project work, current events, nature walks, or your free read. Describe any questions this made you think of.